Perpendicular bisectors

1 Find the equation of the perpendicular bisector of the following line segments:

a A line segment with a gradient of $\frac{1}{2}$ and a midpoint of $(1, 0)$.

b A line segment with a gradient of 2 and a midpoint of $(3, -5)$.

c A line segment with a gradient of -4 and a midpoint of $(7, 9)$.

d A line segment with a gradient of $-\frac{3}{4}$ and a midpoint of $\left(-\frac{1}{2}, 3\right)$.

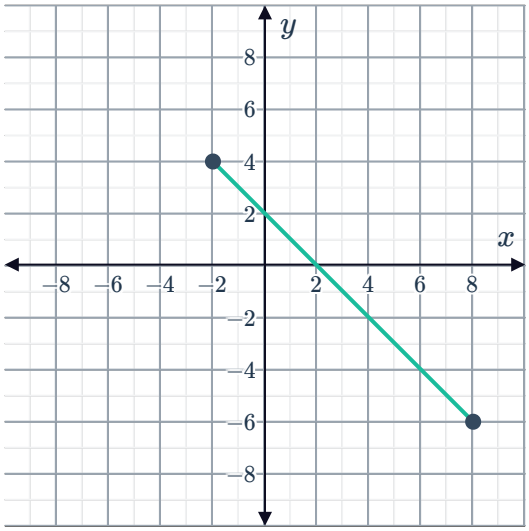
2 For each of the following line segments:

i Find the coordinates of the midpoint of the segment.

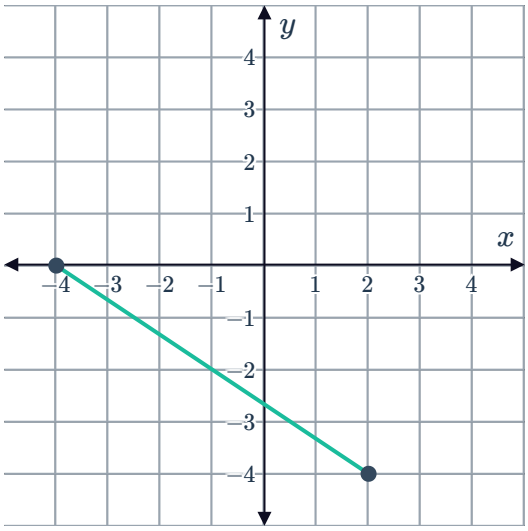
ii Find the gradient of the segment.

iii Find the equation of the perpendicular bisector of the segment.

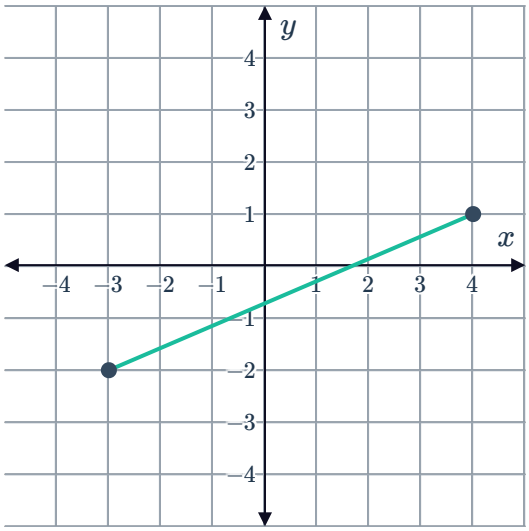
a



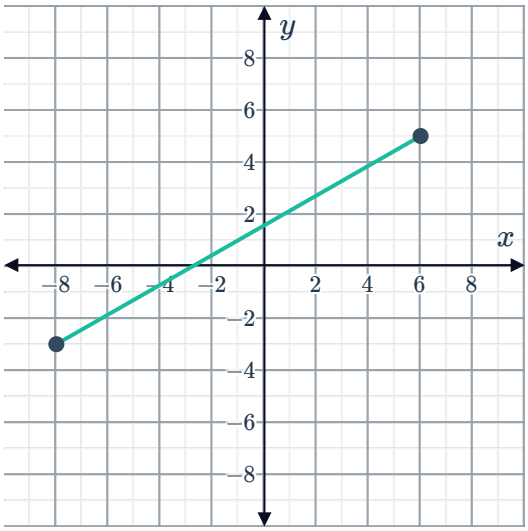
b



c



d



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3 Find the perpendicular bisector of the line segment AB, for each of the following sets of coordinates:

a $A\,(3, 8)$ and $B\,(1, -6)$

b $A\,(7, 11)$ and $B\,(3, -5)$

c $A\,(16, 10)$ and $B\,(14, 20)$

d $A\,(2.5, 6)$ and $B\,(7.5, 12)$

e $A\,(4, 8)$ and $B\,(10, -6)$

f $A\,(-5, -3)$ and $B\,(7, 1)$

g $A\,(10, 0)$ and $B\,(0, 4)$

h $A\,(10, 20)$ and $B\,(30, 60)$

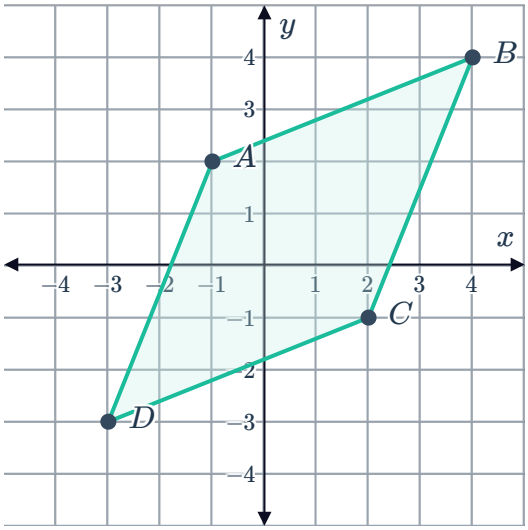
4 A line segment has an equation of $2x + 5y - 8 = 0$, and a midpoint of $(-1, 2)$. Find the equation of the perpendicular bisector of the line segment.

Applications

5 Consider the quadrilateral $ABCD$ shown:

a Find the equation of the perpendicular bisector of the segment AC.

b Determine whether the points B and D lie on this bisector.



6 Three friends live at the following map locations in a town:

- Alvin lives at point $A(4, 6)$
- Bernedette lives at point $B(8, 12)$
- Constantine lives at point $C(14, 2)$

a Find the equation of the perpendicular bisector of:

i AB **ii** BC **iii** AC

b The intersection of these perpendicular bisectors, is the location that is equidistant from all three friends. Use technology to find the coordinates of this map location. Round the values to three significant figures.

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